Dataset Description

Dataset: VinBigData Chest X-ray Abnormalities Detection

1.Dataset Overview:

i. The VinBigData Chest X-ray Abnormalities Detection dataset is a large-scale medical imaging dataset released as part of a Kaggle competition. It contains approximately 18,000 frontal-view chest radiographs collected from multiple medical institutions. The images are stored in DICOM format and are annotated by expert radiologists.
ii. The objective of the dataset is to detect and classify thoracic abnormalities present in chest X-ray images.

2.Dataset Size:

a. Total Images: **18,000**
b. View Type: Postero-Anterior (PA) chest X-rays
c. Each image may contain:
    a. One abnormality
    b. Multiple abnormalities
    c. No abnormality

3.Data Format and Structure

1. Images are stored in DICOM (Digital Imaging and Communications in Medicine) format. DICOM files contain both pixel data and medical metadata.
2. Annotations are stored in a CSV file (train.csv) with the following columns:
    1) image_id – Unique image identifier
    2) class_name – Name of abnormality
    3) class_id – Numeric class label
    4) rad_id – Radiologist identifier
    5) x_min, y_min – Top-left bounding box coordinates
    6) x_max, y_max – Bottom-right bounding box coordinates

4.Target Classes

The dataset includes 14 thoracic abnormality classes and one additional class labelled 'No Finding' to represent the absence of abnormalities.

0– Aortic enlargement
1 -Atelectasis
2 - Calcification
3 -Cardiomegaly
4 -Consolidation
5 -ILD
6 -Infiltration
7 - LungOpacity
8 -Nodule/Mass
9 -Other lesion

10 - Pleural effusion
11 -Pleural thickening
12-Pneumothorax
13- Pulmonary fibrosis
14- No Finding